Project 3: Active Learning for Learning-to-Defer

Human-Centric Artificial Intelligence — experimental report. The system labels AG News articles into four topics (World, Sports, Business, Sci/Tech) and collaborates with a simulated human expert through learning-to-defer.

Task 1 — Baseline classifier

We train a **tfidf** classifier on 20000 labelled training articles. On the held-out AG News test set (7600 articles) it reaches a test accuracy of **90.24%**. This is the baseline the human-AI team must match or beat.

Design choice: TF-IDF + multinomial logistic regression gives calibrated class probabilities cheaply, which the deferral and active-learning stages depend on. A fine-tuned DistilBERT variant shares the same interface and can be swapped in for higher accuracy.

Task 2 — Simulated experts

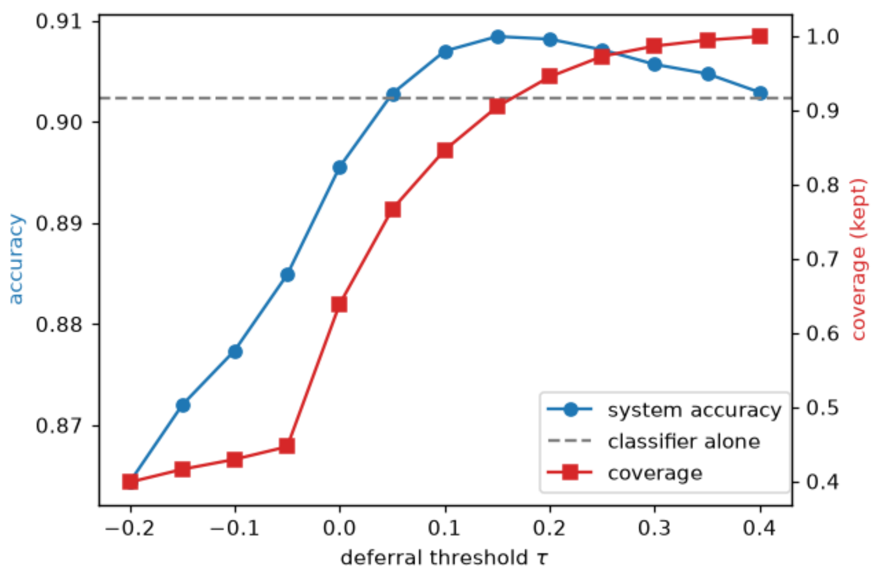
Experts are deliberately imperfect and have *localised* competence: each is reliable on a subset of classes and weak elsewhere. Querying the same example twice returns the same answer, so expert noise cannot be averaged away. Measured accuracies:

Expert	Overall	World	Sports	Business	Sci/Tech
expert_world_sports	0.60	0.96	0.95	0.25	0.25
expert_biz_tech	0.59	0.25	0.25	0.93	0.92

Task 3 — Learning to defer

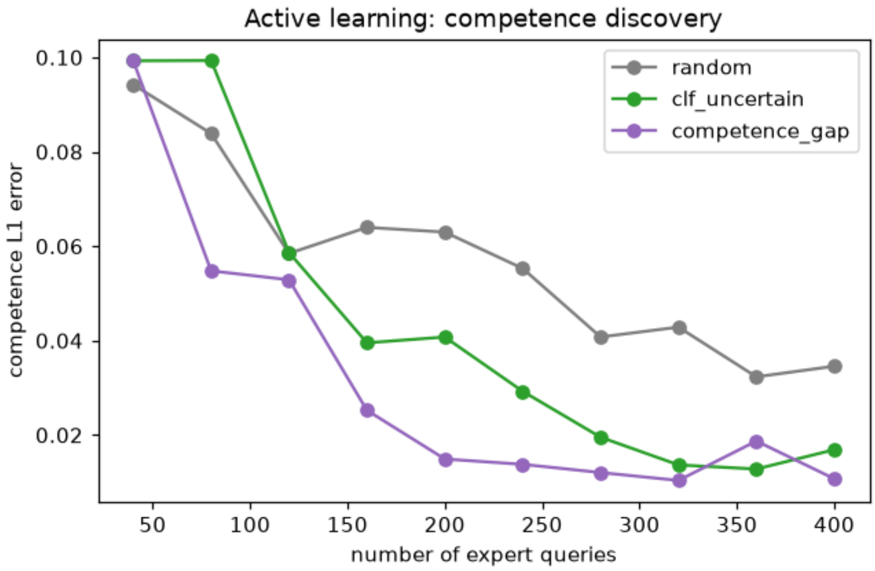
The deferral policy compares, per input, the classifier's max probability against the expert's *expected* correctness (estimated from the predicted-class distribution and a per-class competence vector). It defers when the expert's expected accuracy exceeds the classifier's by a margin τ.

At τ=0.1 the combined system reaches **90.70%** accuracy with 85% coverage, versus 90.24% for the classifier alone and only 59.91% for the expert alone. Selective deferral on the expert's strong classes lifts the team above the baseline.



Task 4 — Active learning for competence discovery

Without any expert labels at training time, we actively query the expert to learn *where* it is competent. Per-class competence is modelled with a Beta posterior; we compare three acquisition strategies. Uncertainty sampling concentrates queries where the classifier is unsure — exactly the region where knowing the expert's skill matters — and recovers the competence profile fastest.



Strategy	Final competence L1 error
random	0.0346
clf_uncertain	0.0169
competence_gap	0.0107

Conclusion

The experiments show a working human-AI team: a strong baseline classifier, imperfect experts with localised skill, a deferral policy that beats both members alone, and an active-learning loop that discovers expert competence from a small query budget.